

The rebalancing engine: one pass per trading day

Signal from yesterday, trade at today's price, pay the cost, record, repeat

1. Observe

BAR t ARRIVES

- Execution prices $S_i(t)$, known at the close
- Signal from closes through $t - 1$: EMAs $\bar{S}_{t-1}^{\text{short}}$, $\bar{S}_{t-1}^{\text{long}}$ give ξ_t
- SIM parameters α_i , β_i and g'_M , held fixed in this lecture

2. Orient

PREFERENCES

Preference weight of firm i :

$$\gamma_i(t) = \tanh\left(\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t} g'_M\right)$$

- sign: $\gamma_i > 0$ is preferred ($\mathcal{A}^+$), otherwise non-preferred ($\mathcal{A}^-$)
- magnitude: share of the budget

3. Decide

TARGETS AND RULES

Cobb–Douglas targets on $\mathcal{A}^+$:

$$n_i^* = \frac{\gamma_i}{\sum_{j \in \mathcal{A}^+} \gamma_j} \frac{W_{\text{adj}}}{S_i(t)}$$

- schedule: trade only if $t \in \mathcal{R}$
- turnover cap $\tau_{\max}$ scales the move
- drawdown limit $d_{\max}$: go to cash

4. Act

TRADE AND RECORD

- Trade Δn_i at $S_i(t)$
- Pay $C_t = c \sum_i |\Delta n_i| S_i(t)$ before the targets are sized
- $W_{\mathcal{P}}^+(t) = W_{\mathcal{P}}^-(t) - C_t$; update the cash
- Record positions, weights, cost

next bar $t + 1$: cash accrues $e^{g_f \Delta t}$, mark to market at $S_i(t + 1)$, check the drawdown breaker

A static allocation (L6a, L7a) is one pass on the decision date, then hold.